

Speaker text: Does conditioning leave a structural memory?

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Spoken words, 21 main frames: **2083**; read aloud at 130 words per minute: **16.0 min**. Gray text in brackets is optional or for questions only and is not counted.

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Today we ask whether conditioning leaves a structural memory in the metal.

Last year at MeVArc we showed our dislocation model and STEM cross-sections of a modified near-surface layer, and we closed by asking for more comparative forensics.

This talk brings the large-area answer on copper, and a first look at stainless steel.

We start with what conditioning does.

Conditioning works, but we do not know what it changes in the metal.

Every high-gradient device reaches its usable field by conditioning.

The breakdown rate rises roughly as the thirtieth power of the field.

Conditioning tracks the number of high-field pulses, not the number of breakdowns, so the field, more than the arcs, drives it.

The curve is a schematic: field holding climbs over about a billion pulses.

Something in the metal stores that improvement.

[optional: Walter set out these clues in his invited talk this morning.]

Our answer starts from a model.

3 / 21 Our model: breakdown as collective dislocation dynamics

92 words

Our model treats breakdown as collective dislocation dynamics.

Field holding rises from FCC to BCC to HCP metals, matching the order of barriers to dislocation motion.

In MDDF, the mobile dislocations driven by the Maxwell stress fluctuate, and breakdown nucleates when that population runs away.

We fitted the same model to the thirtieth-power rate and its temperature dependence, and it then reproduced the histogram of dark-current spikes, which was a prediction.

Its central prediction for this talk is that conditioning evolves the near-surface dislocation structure.

Next: what that means for a microscope.

This slide turns the model into something a microscope can look for.

The mobile population is fast: it fluctuates and decays back.

Conditioning means fewer breakdowns at the same field, so the barrier, the hill in this picture, must rise.

A memory that lasts about a billion pulses must then sit in the slow, stored dislocations that set the hill, and those bend the lattice, which EBSD reads.

The field writes the memory, so we look far from craters and expect the order to follow the local field.

The model predicts where and in which order; it does not predict size, depth or sign.

Last year's STEM gave a first look.

STEM has already imaged an altered near-surface dislocation structure, but only at specific locations.

These are our 2025 cross-sections of conditioned hard copper, imaged at HUJI by Inna Popov.

Under the protective coating, the top 200 nanometers or so of the field-exposed region is nearly free of dislocation walls.

We see fewer walls at 300 and at 30 kelvin, and the denuded layer is clearest at 30 kelvin.

This fits an independent clue: soft copper conditions far more slowly than hard copper, so the starting population matters.

Two cautions: FIB preparation plants its own dislocations, so we trust only large-scale structure, and one cross-section cannot speak for the whole electrode.

What was missing was the large-area view.

[if asked: the paper gives no thickness; about 200 nanometers is read from its Figure 10. Hard copper conditioned within about 25 million pulses; soft copper needed some 600 to 900 million.]

The load is tiny, but it is repeated about a billion times.

The Maxwell stress at 80 megavolts per meter is 0.028 megapascals, about a thousand times below the yield of annealed copper.

Fatigue at ten to the eight to ten to the ten cycles rearranges dislocations irreversibly well below yield, which is our regime.

Pulsed DC means no RF pulsed heating and no bulk currents, so the Maxwell stress is the dominant cyclic load.

MDDF needs only the stress that moves mobile dislocations in a frustrated crystal, far below yield: a dense, pinned network in which weak internal stresses, not the applied one, decide where mobile segments move.

[optional: Jianyu, with Guodong Meng, and Veronika show field-driven deformation at the nanoscale this afternoon, at 16:00 and 16:30.]

So the model predicts a large-area signature: is it there?

Last year we asked for more comparative forensics, and this slide shows what came back.

On the left is what we showed in 2025: the model, and local STEM of the denuded zone, from a FIB lamella overview to a cross-section at the surface.

We closed with two questions: other materials, and more samples.

On the right is the large-area answer, an EBSD survey across one conditioned cathode.

Let us look at how that cathode was exposed.

8 / 21 One cathode turns field exposure into a controlled coordinate

115 words

One cathode gives us a controlled range of field exposure.

The anode is sloped, so radius maps onto local field: about 80 megavolts per meter under the flat center, 73 to 74 at the edge, and below about 2.5 beyond the anode radius.

The cathode is heat-treated OFE copper, conditioned with 1-microsecond pulses at 1 kilohertz, about ten to the nine pulses.

We mapped nine EBSD regions: center, edge, periphery, and an identically treated reference that never saw field.

Copper for high field, purple for the periphery and teal for the reference stay the same on every data slide.

Comparing regions on one specimen removes cathode-to-cathode differences.

[optional: this is the CERN pulsed-DC system; Victoria gave its update this morning.]

Here is what EBSD sees on that surface.

We use EBSD because it covers a large area far from breakdown craters.

The copper box is one 500-micrometer map; the dashed circles are craters, the nearest about half a millimeter away.

The 3-micrometer step sets only the lateral resolution; the signal comes from the top 40 nanometers of copper at 15 kilovolts, so this is a near-surface measurement.

Each region has about 24,000 indexed points.

Our main metric is LAM, the local average misorientation, with KAM and LOS as cross-checks, and misorientation is a proxy for the density of geometrically necessary dislocations.

Our target is the cumulative response to the field, not crater damage.

Next: what one pixel measures.

[if asked: a shallow FIB clean removes the oxide, under 10 nanometers by AFM, with the same protocol on all nine regions.]

Every colored pixel on the next slides is one angle, read from the top 40 nanometers.

The 15-kilovolt beam visits the surface point by point on a 3-micrometer grid.

Each point gives a Kikuchi pattern, and indexing turns it into one lattice orientation.

LAM averages the rotation to the six neighbors, dropping pairs above five degrees, so grain boundaries are excluded.

On this scale the reference sits at 0.68 degrees and high field at about 1.2.

A bent lattice needs geometrically necessary dislocations, so LAM is a dislocation proxy over half a millimeter, not a dislocation image.

Next: how to read a LAM map.

The color scale decides what a LAM map shows.

On a zero-to-50-degree scale, only grain boundaries stand out.

Rescaling to zero to five degrees reveals the curvature inside the grains, the low-angle misorientation that tracks geometrically necessary dislocations.

That intragrain signal is what we care about.

Every map from here on uses the same zero-to-five-degree scale.

Now we compare high field with the reference.

High-field copper carries more curvature inside its grains.

Both maps use the same color scale and the same magnification, cropped to the same field of view.

Look at the texture, not only the average color: the field-exposed grains are full of green and yellow, while the reference stays mostly blue.

This pair is representative; the numbers come from all nine regions.

Here are those numbers.

The signal separates into three tiers, and the tiers follow field exposure.

Center and edge, both at high field, sit at about 1.2 degrees, and the external reference at about 0.68, so high field is about 75 percent higher.

The periphery, on the same cathode, saw at most 2.5 megavolts per meter and sits in between.

It shared machining, heat treatment, mounting, vacuum and handling with the high-field regions, so it is a low-field internal control.

Grain size does not follow this hierarchy; the ordering follows field history.

The 75 percent describes this cathode and its history, not a universal constant.

Next, the shape of the distributions.

[if asked: breakdown density falls from about 24 to 12 to 5 per square centimeter from center outward; center and edge differ twofold in breakdown density but not in LAM.]

Field exposure does more than shift the distribution: it builds a high-misorientation tail.

The dotted line marks two degrees, and the shaded areas are the tails beyond it.

The fraction above two degrees is about 0.14 at high field against 0.016 on the reference, roughly eightfold.

A gamma fit keeps its shape, k near 2.7, while the scale roughly doubles.

The whole distribution stretches, and a uniform offset cannot do that.

LAM, KAM and LOS all agree.

So what does this say about the model?

[if asked: KS, chi-squared and Anderson–Darling tests all separate the distributions; we quote effect sizes and grain-corrected errors rather than pixel-level p-values.]

The large-area test supports the model's prediction.

To our knowledge, this is the first large-area observation that conditioned regions carry a different near-surface dislocation structure, ordered as MDDF predicts and following the simulated local field.

The cartoon is our interpretation from the manuscript: one load acting on two starting populations.

Dense, as-machined copper loses dislocations near the surface, the layer STEM saw; heat-treated copper has little to lose, so the same load builds the curvature EBSD records.

Size, depth and sign were not predicted.

Which of the stored dislocations matter?

Those near enough to the surface to exchange stress with it, and that is the next slide.

[if asked, the theory–experiment record: predicted observable, a change in the stored near-surface population; measured, LAM; mapping, a relative GND proxy under one protocol; no parameters fitted to the EBSD data; no prediction of the size of the change.]

[if asked about E_S : its simulated profile has the same order; a candidate basis, not a measurement.]

16 / 21 Elastic screening: a sheath of dislocations at the surface

139 words

Three length scales decide what the surface sees; in the plasma on the left, only a sheath one Debye length thick reaches the wall.

First, the field stops within 0.1 nanometers.

Second, weak internal fluctuations are screened within about 25 nanometers, and only that layer, which EBSD samples, talks to the surface.

Third, the applied load is not screened, since dislocation stress is self-equilibrated; it reaches millimeters.

Copper is a plasma of dipoles, Debye-like for fluctuations and dielectric-like for the mean load: it reduces, it does not nullify; and the surface is an absorbing wall: a sheath of one screening length sets what it sees, hence emission and breakdown.

In our picture, weak screened internal stresses set the structure, and the strong unscreened load drives the mobile dislocations through it.

Next: what copper leaves open, and where we go.

[two time constants: think of the mobile dislocations as the electrons and the stored network as the ions. The one-microsecond pulse is far above the network's plasma frequency, so it passes unscreened; the mobile population responds within the pulse and screens the slow internal fluctuations. Ten-to-the-nine repetitions act like a ponderomotive force—a slow, cumulative, nonlinear push of a fast drive—and the Maxwell stress itself is the ponderomotive force of the field on the metal.]

[if asked why conditioning does not need to screen the applied stress: it cannot, and it need not. What changes is the medium's response function—stored density, sources, ℓ_D —i.e. the threshold of the mobile-population instability, not the drive.]

[if asked about the dielectric analogy of earlier versions: dislocations are dipoles, not free monopoles, so the mean load is reduced, never cancelled.]

[if asked: the 25 nanometers is an order-of-magnitude estimate, Debye-like screening in a two-dimensional single-slip dislocation gas (Groma 2006).]

Copper shows the effect; it does not show universality.

What is shown: a large-area structural difference, three metrics that agree, internal and external controls, and support for the MDDF prediction.

What remains open: one conditioned cathode in one experiment, no full depth profile, no grain-orientation dependence yet, no further cathodes and no second material.

On this cathode, field level and breakdown density vary together, and LAM is a relative proxy, compared under one protocol.

A stronger test changes both the depth we probe and the material.

That is where we go next.

Our next steps run along two axes, depth and material.

Today's result sits at about 40 nanometers in heat-treated copper; the 2025 STEM depletion in hard copper was about 200 nanometers deep.

Going deeper means cross-sections from a mapped high-field region and from the reference, through the top micrometer.

They ask whether the EBSD layer and the denuded zone belong to one depth profile, and whether the denuded zone is the sheath, about eight screening lengths wide.

We carry over the artifact controls from the 2025 red herrings.

Depth continues our own STEM line, published with Uppsala.

Along the material axis comes stainless steel under continuous DC, with Consorzio RFX, on the next slide.

Steel is a different material, so we expect different structural evolution.

Lower stacking-fault energy means planar slip, and continuous DC gives a steady traction of about 16 kilopascals not a billion pulses.

Its metastable austenite is a built-in strain gauge: epsilon and alpha-prime martensite mark plastic strain above a few tenths of a percent.

With Consorzio RFX, Antonio De Lorenzi, Nicola Pilan and Emanuele Spada, two 304L electrodes ran 90 and 73 hours at 59 to 61 megavolts per meter, zero breakdowns.

On 26 EBSD maps the exposed apex lies inside or below the unexposed range, and with two to four maps per group a field effect cannot yet be resolved.

Copper supports the memory; steel asks whether it is universal.

Nicola will tell you about the RFX program next.

[if asked: we have not yet looked for martensite on an exposed surface. Next on steel: one electrode mapped before and after being driven into breakdowns, with epsilon and alpha-prime in the phase list, so that absent martensite becomes a strain bound. 80 percent power at one standard deviation needs about 17 maps per group.]

Conditioning appears to leave a subsurface memory.

Our model predicts that conditioning evolves the near-surface dislocation structure.

STEM saw it locally in 2025, and large-area EBSD now supports it.

The four bars are the tier means: 1.21 and 1.19 degrees at center and edge, 0.79 at the periphery and 0.68 on the reference, about 75 percent higher at high field.

Depth and stainless steel now test whether the memory survives a change of scale, alloy and loading.

The paper is on arXiv, 2606.19192.

To close, what we found and what we are asking for.

Three conclusions, and an invitation.

Conditioning leaves a structural memory in the near-surface dislocation population; it is supported on one cathode and not yet replicated.

In our picture it lives within a screening length of the surface, the layer EBSD sees and the layer that sets emission.

What would settle it is the same area mapped before and after, a depth profile, and a second material, steel with Consorzio RFX.

So talk to me about new materials, other electrode configurations and same-area mapping, and the QR code opens the paper.

[optional: Mircea's resistivity measurements, at 11:30 just before us, probe the same state.]

Thank you.

Nicola will tell you about the RFX program next.

This is the field-exposed versus reference pair from the 300-kelvin cathode.

On the field-exposed side the dislocation walls are strongly reduced, while the reference keeps walls up to the surface.

In the published paper this pair is Figure 9; the zone the paper calls denuded, and shows most clearly, is on the 30-kelvin cathode, Figure 10.

The paper states no thickness.

Back to the questions.

[if asked: the 200 nanometers is read from the Figure 10 scale bar.]